

INFO 510 – Bayesian Modelling and Inference

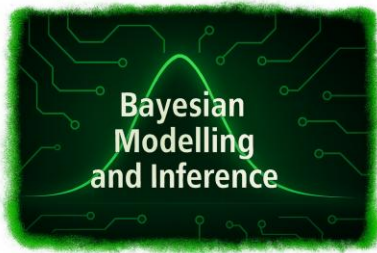
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Final Project Proposal

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Introduction (project statement and goals)

Automatic music genre classification is a key challenge in music information retrieval, demanding models that capture rhythmic, harmonic, and timbral structures across time. Using the GTZAN dataset—1,000 thirty-second audio clips across 10 genres such as classical, jazz, pop, and rock—each track is represented as a spectrogram that visualizes sound energy over time and serves as the input to a deep neural network.

This project, **B.E.A.R. D.O.W.N. — Bayesian Evaluation of Audio Representations: A Deep Optimized Waveform Network**, aims to develop a Bayesian Deep Neural Network (BDNN) that achieves 80% genre-recognition accuracy while providing calibrated uncertainty estimates. The study addresses **three goals**:

1. Determine whether a BDNN can accurately classify music genres while quantifying predictive uncertainty.
2. Apply Bayesian inference to the network's learned embeddings to explain which spectral features drive genre distinctions and confusions.
3. **Stretch Goal:** Explore whether predictive uncertainty can guide *active learning*—selecting ambiguous or underrepresented spectrograms for retraining—to further improve genre classification performance and reduce generalization error.

Together, these goals integrate deep representation learning with probabilistic reasoning to build a model that both classifies and explains musical style with measurable confidence.

Dataset – Selection and provenance

The dataset can be found at:

<https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification>

The original dataset consists of:

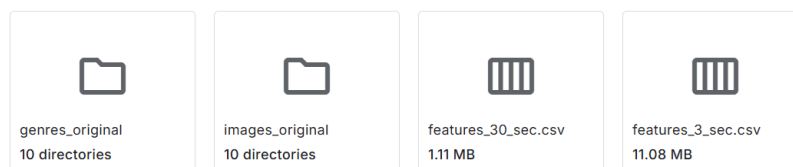


Figure 1. Topology of the GTZAN dataset.

The dataset is composed of audio directories and accompanying feature files: *genres_original* (10 genre folders of 30-second audio clips), *images_original* (10 directories of color spectrograms), and two CSV files containing pre-computed audio features for the 30-second and 3-second segments.

Note — *the images_original directory contains color spectrogram representations of each track.*

Gen ai statement

chatPGT was used to make HTML code for nicer formatting of the text and grammar checks, and MS Copilot for image generation.

EDA plan

EDA will consist of checking for – (1) complete data set, (2) complete feature set.

Questions addressed

Predictive Question (Q1):

Can a Bayesian Deep Neural Network (CNN backbone + variational Bayesian classifier head) accurately classify GTZAN music genres while providing calibrated predictive uncertainty? (goal of at least 80%, with a generalization gap [the gap between the training and validation sets] to be less than 5%.)

Inferential Question (Q2):

Using the latent embeddings learned by the network, can a Bayesian regression model (via MCMC) identify which audio features most strongly drive genre distinctions and misclassifications?

Active Learning Question (Q3 – Stretch Goal):

Can the predictive uncertainty estimates from the Bayesian Deep Neural Network be leveraged to guide active learning—selecting ambiguous or underrepresented spectrograms (e.g., those with high predictive entropy, small confidence margins, or high mutual information across Monte Carlo samples)—to further improve genre classification accuracy and reduce generalization error?

Tentative plan of analysis (Q1 and Q2 only)

Model Architecture (Q1):

- Input: 128×128 grayscale spectrograms.
- CNN backbone: 4 convolutional blocks + global average pool $\rightarrow$ 256-D embedding.
- Bayesian head: two variational linear layers ($256 \rightarrow 128 \rightarrow 10$).
- Priors: independent Gaussian priors $W, b \sim \mathcal{N}(0, 1^2)$.
- Inference: variational Bayes (ELBO = cross-entropy + β ·KL); training with AdamW optimizer; Monte-Carlo weight sampling for posterior predictive checks.
- Validation: 6-fold cross-validation; metrics = accuracy, F1, ECE (expected calibration error), and Brier score; reliability diagrams for calibration assessment.

Statistical Model (Q2):

- Extract 256-D embeddings (or PCA-reduced to 10–20 D) from the trained BDNN.
- Fit a **Bayesian multinomial logistic regression** using *Stan* (or *PyMC*) with priors $\beta_j \sim \mathcal{N}(0, 1)$, hierarchical genre-level intercepts $\alpha_g \sim \mathcal{N}(0, 1)$.
- Inference: Hamiltonian Monte Carlo (NUTS), 4 chains $\times$ 2 000 iterations (half warm-up).
- Diagnostics: trace plots, $\hat{R} < 1.01$, effective sample size (ESS), and posterior predictive checks (PPC).
- Validation: held-out embeddings to evaluate log-loss and posterior predictive accuracy.

Software: PyTorch for the BDNN, PyMC/Stan for MCMC, Matplotlib + Seaborn for visualizations.

Expected outcomes

The **Bayesian Deep Neural Network (BDNN)** is expected to achieve an overall **test accuracy of approximately 80%** on the GTZAN dataset, with **validation accuracy within ± 2 –3%** of that value—indicating **good generalization** and the absence of significant overfitting. The model should maintain a **balanced precision–recall trade-off** across genres (macro-F1 ≥ 0.78) and exhibit **well-calibrated posterior probabilities** (expected calibration error < 0.05). Posterior predictive variance is anticipated to remain **low for distinct genres** (e.g., *classical*, *metal*) and **higher for acoustically overlapping categories** (e.g., *blues* vs. *country*), reflecting uncertainty where class boundaries blur. The accompanying **Bayesian logistic regression** on the learned embeddings should highlight **credible feature–genre relationships** that clarify both correct and ambiguous classifications. Collectively, these outcomes will demonstrate that **Bayesian deep learning can yield reliable, interpretable predictions with calibrated confidence**, supporting its integration into **robust, generalizable music-recommendation systems**.